

HillSmart

Routes that understand hills, stairs, and effort, with hands-free voice guidance.

The Idea

Ithaca is famous for its hills. Normal maps pick the shortest route, even if it climbs Libe Slope or uses stairs. HillSmart lets you choose what matters: *least climbing*, *no stairs* (wheelchair, stroller, scooter, bike), or *fastest*. It shows the elevation profile of each option and guides you hands-free by voice, which is useful on a bike or with a wheelchair.

Track Fit

Built with Cursor	Routing, elevation analysis, and UI built in Cursor.
Grok Voice / Imagine	Voice turn-by-turn and questions (“ <i>Is there a flatter way?</i> ”).
Space data	Not needed. (Optional twist: elevation data comes from satellite measurements.)
Navigation theme	Direct fit; accessibility angle.
Also eligible	Big Red, Software, Design.

Features

MVP

- Enter start and destination (by voice or text).
- Get 2–3 candidate routes; compute total climb and steepest section for each.
- Elevation profile chart and a map with the route color-coded by steepness.
- Voice summary: “*Route B adds 3 minutes but saves 40 meters of climbing.*”

Stretch

- Custom routing that truly avoids steep segments (not just picking between options).
- Crowd-reported obstacles (broken elevator, icy path).
- Live voice turn-by-turn while moving.

Tech Stack and Data

- Routes: Google Maps Directions API (alternatives) or OpenStreetMap with OSRM/GraphHopper.
- Elevation: Google Elevation API or open elevation data.
- Frontend: React with Leaflet/Google Maps; a chart library for profiles.
- AI: Grok Voice API.

Limitations and Risks

- **Picking vs. planning.** Choosing among Google’s alternative routes is easy but limited; real slope-aware routing needs your own routing engine.
Mitigation: Ship “compare alternatives” first; custom routing is a stretch goal.
- **Stairs data is incomplete.** OpenStreetMap marks many but not all stairs and ramps on campus.
Mitigation: Hand-check the demo routes and add crowd reports.
- **Elevation resolution.** Public elevation data can miss short steep sections and stairs.
Mitigation: Treat numbers as estimates; combine with stairs tags.
- **Existing features.** Google Maps shows wheelchair-accessible transit and bike elevation in some places.
Mitigation: Focus on walking/wheelchair hill avoidance on campus, which Google doesn’t do well.
- **API cost and keys.** Google Maps APIs need billing set up.
Mitigation: Use free credits or OpenStreetMap; never commit the key.

Scorecard

Demo-able	4/5	Clear map demo; anyone at Cornell relates.
Buildable in 24h	4/5	MVP is mostly APIs.
Navigation theme	5/5	Direct fit.
Wow factor	3/5	Useful, not flashy.
Technical risk (5 = riskiest)	2/5	Low (MVP), moderate (custom routing).

Verdict

Good practical pick. Very relatable to Cornell judges. The voice part must feel essential, not added on.